Cambridge International AS & A Level

Cambridge Assessment International Education
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MATHEMATICS **9709/63**

Paper 6 **October/November 2018**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

- The symbol FT implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.

  - Note: B2 or A2 means that the candidate can earn 2 or 0.
    B2/1/0 means that the candidate can earn anything from 0 to 2.

  The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.

- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking *g* equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only – often written by a ‘fortuitous’ answer
ISW	Ignore Subsequent Working
SOI	Seen or implied
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1 A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.

PA –1 This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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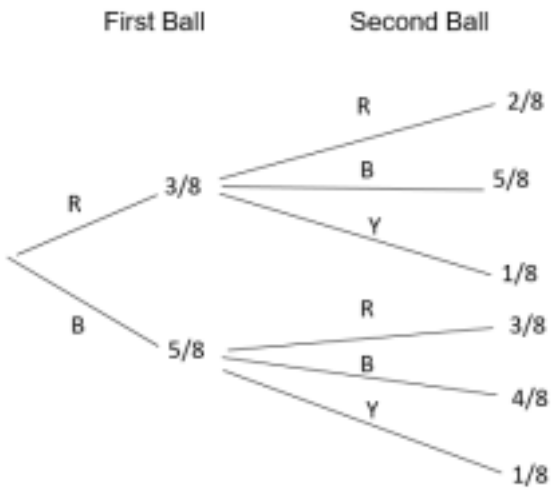
Question	Answer	Marks	Guidance
1	**Method 1**		
	... M … M … M … M … M …	**M1**	k × 5! (120) or k × 6P2 (30), k is an integer ⩾ 1,
	No. ways men placed × No. ways women placed in gaps = 5! × $^6P_2$	**M1**	Correct unsimplified expression
	= 3600	**A1**	Correct answer
	Method 2		
	Number with women together = 6! × 2 (1440) Total number of arrangements = 7! (5040)	**M1**	6! × 2 or 7! – k seen, k is an integer ⩾ 1
	Number with women not together = 7! – 6! × 2	**M1**	Correct unsimplified expression
	= 3600	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
2(i)	x: −2, −1, 0, 1, 2, 3 P($X = x$): $\frac{2}{18}$, $\frac{4}{18}$, $\frac{5}{18}$, $\frac{4}{18}$, $\frac{2}{18}$, $\frac{1}{18}$	**B1**	–2, –1, 0, 1, 2, 3 seen as top line of a pdf table OR attempting to evaluate P(–2), P(–1), P(0), P(1), P(2), P(3),
		B1	At least 4 probs correct (need not be in table)
		B1	All probs correct in a table
		3	

Question	Answer	Marks	Guidance
2(ii)	$\mathrm{E}(X) = \dfrac{-4-4+0+4+4+3}{18} = \dfrac{1}{6}$	**M1**	Correct unsimplified expression for the mean using their table, $\Sigma p = 1$, may be implied
	$\mathrm{Var}(X) = \dfrac{8+4+0+4+8+9}{18} - \left(\dfrac{1}{6}\right)^2$ $=11/6 - 1/36\ (1.8333 - 0.02778)$	**M1**	Correct, unsimplified expression for the variance using their table, and their mean^2 subtracted. Allow $\Sigma p \neq 1$
	$= 65/36$, (1.81)	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
3(i)		**B1**	Fully correct labelled tree and correct probabilities for 'First Ball'
		B1	Correct probabilities (with corresponding labels) for 'Second Ball'
		2	
3(ii)	P(RR) + P(BB) = 3/8 × 2/8 + 5/8 × 4/8 = 3/32 + 5/16	**M1**	Correct unsimplified expression from their tree diagram, Σp = 1 on each branch
	= 13/32 (0.406)	**A1**	Correct answer
		2	

Question	Answer	Marks	Guidance
3(iii)	P(RB) = $3/8 \times 5/8$ = 15/64	**M1**	$P(\text{1st ball red}) \times P(\text{2nd ball blue})$ from their tree diagram seen unsimplified as numerator or denominator of a fraction Allow $\Sigma p \neq 1$ on each branch
	P(B) = 3/8 × 5/8 + 5/8 × 4/8 = 35/64	**M1**	Correct unsimplified expression for P(B) from their tree diagram seen as denominator of a fraction. Allow $\Sigma p \neq 1$ on each branch
	P(R|B) = P(RB) / P(B) = (15/64) ÷ (35/64) = 3/7 (0.429)	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
4(i)	Total number of selections = ${}^{12}C_7$ = 792	**B1**	Seen as denominator of fraction
	Selections with boy included = ${}^{11}C_6$ or ${}^{12}C_7 - {}^{11}C_7$ = 462	**M1**	Correct unsimplified expression for selections with boy included seen as numerator of fraction
	Probability = 462/792 = 7/12 (0.583)	**A1**	Correct answer
	OR		
	prob of boy not included = 11/12 × 10/11 × …. × 5/6= 5/12	**B1**	Correct unsimplified prob
	1 – 5/12	**M1**	Subtracting prob from 1
	= 7/12	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
4(ii)	**Method 1**		
	Scenarios are: 2G + 5B: $^4C_2 \times {}^8C_5 = 336$	**B1**	One unsimplified product correct
	3G + 4B: $^4C_3 \times {}^8C_4 = 280$ 4G + 3B: $^4C_4 \times {}^8C_3 = 56$	**M1**	No of selections (products of ^nC_r and ^nP_r) added for 2, 3 and 4 girls with no of girls and no of boys summing to 7
	Total = 672	**A1**	Correct total
	Probability = 672/792 (28/33) (0.848)	**A1ft**	Correct answer – 'total'/('total no of selections' from i)
	Method 2		
	0G + 7B $^4C_0 \times {}^8C_7 = 8$	**B1**	One unsimplified no of selections correct
	1G + 6B $^4C_1 \times {}^8C_6 = 112$ Total = 8 + 112 = 120	**M1**	No of selections (products of ^nC_r and ^nP_r) added for 0 and 1 girls with no of girls and no of boys summing to 7
	$(^{12}C_7 - 120)/792$ or $1 - 120/792$	**A1**	792 – 120 = 672 or 1 – 120/792
	Probability = 672/792 (28/33) (0.848)	**A1ft**	'672' over '792' from i
	Method 3 (probability)		
	1 – P(0) – P(1) = 1 – (8/12 × 7/11 × …… × 2/6) – (8/12 × … × 3/7 × 4/6 × 7)	**B1**	One correct unsimplified prob for 0 or 1
	= 1 – 1/99 –14/99	**M1**	Subtracting 'P(0)' and 'P(1)' (using products of 7 fractions with denominators from 12 to 6) from 1
		A1	Both probs correct unsimplified
	= 84/99 = 28/33	**A1ft**	1 – 'P(0)' – 'P(1)'

Question	Answer	Marks	Guidance
4(ii)	**Method 4 (probability)**		
	P(2) + P(3) + P(4) =	**B1**	One correct unsimplified prob for 2, 3 or 4
	42/99 + 35/99 + 7/99	**M1**	Adding 'P(2)', 'P(3)' and P(4)' (using products of 7 fractions with denominators from 12 to 6)
		A1	Three probs correct unsimplified
	= 84/99 = 28/33	**A1ft**	'P(2)'+ 'P(3)' + 'P(4)'
		4	

Question	Answer	Marks	Guidance
5(i)	$z_1 = \pm\frac{90-120}{24} = -\frac{5}{4}$, $z_2 = \pm\frac{140-120}{24} = \frac{5}{6}$	**M1**	At least one standardisation, no cc, no sq rt, no sq using 120 and 24 and either 90 or 140
	$= \Phi\left(\frac{20}{24}\right) - \Phi\left(-\frac{30}{24}\right)$	**A1**	–5/4 and 5/6 unsimplified
	$= \Phi(0.8333) - (1 - \Phi(1.25))$ = 0.7975 – (1 – 0.8944) or 0.8944 – 0.2025 = 0.6919	**M1**	Correct area $\Phi - \Phi$ legitimately obtained and evaluated from phi(their z_2) – phi (their z_1)
	= 0.692 AG	**A1**	Correct answer obtained from 0.7975 and 0.1056 oe to 4sf or 0.6919 seen www
		4	

Question	Answer	Marks	Guidance
5(ii)	**Method 1**		
	Probability = P(2, 3, 4) $= 0.692^2(1 - 0.692)^2 \times {}^4C_2 + 0.692^3(1 - 0.692) \times {}^4C_3 + 0.692^4$	**M1**	Any binomial term of form $4C_x p^x (1-p)^{4-x}$, $x \neq 0$ or 4
		B1	One correct bin term with $n = 4$ and $p = 0.692$,
	= 0.27256 + 0.40825 + 0.22931	**M1**	Correct unsimplified expression using 0.692 or better
	= 0.910	**A1**	Correct answer
	Method 2:		
	1 – P(0, 1) =	**M1**	Any binomial term of form $4C_x p^x (1-p)^{4-x}$, $x \neq 0$ or 4
	$1 - 0.692^0(1 - 0.692)^4 \times {}^4C_0 - 0.692^1(1 - 0.692)^3 \times {}^4C_1$	**B1**	One correct bin term with $n = 4$ and $p = 0.692$
	= 1 – 0.00899 – 0.0808757	**M1**	Correct unsimplified expression using 0.692 or better
	= 0.910	**A1**	Correct answer
		4	

Question	Answer	Marks	Guidance
6(i)	P(X >1800) = 0.96, so P($Z > \frac{1800-2000}{\sigma}$) = 0.96	**B1**	± 1.75 seen
	$\Phi(\frac{200}{\sigma}) = 0.96$ $\frac{200}{\sigma} = 1.751$	**M1**	$z = \pm\frac{1800-2000}{\sigma}$, allow cc, allow sq rt, allow sq equated to a z-value
	$\sigma = 114$	**A1**	Correct final answer www
		3	
6(ii)	Mean = 300 × 0.2 = 60 and variance = 300 × 0.2 × 0.8 = 48	**B1**	Correct unsimplified mean and variance
	P(X < 70) = P($Z > \frac{69.5-60}{\sqrt{48}}$)	**M1**	$Z = \pm\frac{x - their\,60}{\sqrt{their\,48}}$
	$= \Phi(1.371)$	**M1**	69.5 or 70.5 seen in an attempted standardisation expression as cc
	=0.915	**A1**	Correct final answer
		4	
6(iii)	np = 60, nq = 240: both > 5, (so normal approximation holds)	**B1**	Both parts evaluated are required
		1	

Question	Answer	Marks	Guidance
7(i)	Anvils | | Brecons 8 | 15 | 9 5 | 16 | 6 5 3 2 0 | 17 | 0 1 2 2 8 4 1 0 | 18 | 1 2 3 3 6 | 19 | 2 Key: 5|16|6 means 165 cm for Anvils and 166 cm for Brecons	**B1**	Correct stem, up or down
		B1	Correct Anvils labelled on left, leaves in order from right to left and lined up vertically, no commas
		B1	Correct Brecons labelled on same diagram on right hand side in order from left to right and lined up vertically, no commas
		B1	Correct key, not split, both teams, at least one with cm
		4	
7(ii)	Median = 173	**B1**	Correct median (or Q2)
	LQ = 169; UQ = 181 IQR = 181 – 169	**M1**	Either UQ = 181 ± 4, or LQ = 169 ± 4 and evaluating UQ – LQ
	= 12	**A1**	Correct answer from 181 and 169 only
		3	

Question	Answer	Marks	Guidance
7(iii)	$\Sigma x = 1923 + 166 + 172 + 182$ (= 2443) $\sum x^2 = 337221 + 166^2 + 172^2 + 182^2$ (= 427485)	**M1**	Correct unsimplified expression for $\sum x$ and $\sum x^2$, may be implied
	Mean = $\frac{\sum x}{14} = \frac{2443}{14} = 174.5$	**M1**	Correct unsimplified mean
	Variance = $\frac{\sum x^2}{14} - \left(\frac{\sum x}{14}\right)^2 = \frac{427485}{14} - \left(\frac{2443}{14}\right)^2$	**M1**	Correct unsimplified variance using 14, their Σx and their Σx^2, not using 1923 and/or 337221
	S d = 9.19	**A1**	Correct answer
		4	